
Software Requirements Specification for a Personal Media Content Tracking Information System

Software Requirements Specification

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Abstract

This document is a Software Requirements Specification (SRS) for the personal media content tracking web application covering books, TV series, and films. The system is intended for registered users who maintain a personal catalog of viewed and read content. The document defines functional requirements (31 requirements), non-functional requirements (9 requirements), acceptance criteria, BDD scenarios, and a traceability matrix for version v1 (MVP). The document was prepared using the SKED methodology (Socratic Knowledge Elicitation & Documentation) as part of the “Information Systems Design” course. The MVP includes: registration and authentication, a catalog with CRUD operations, tags with autocomplete, a Sessions mechanism for recording repeated viewings, collections (many-to-many), automatic completion date fill, and personal statistics. Social interaction features, offline mode, and integration with third-party media databases are deferred to future versions.

Keywords: software requirements specification, media tracker, web application, SRS, content catalog, Socratic dialogue

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1. Introduction

1.1. Purpose and Scope of the Document

This document is a Software Requirements Specification (SRS) for the personal media content tracking information system. The purpose of this document is to formally describe the functional and non-functional requirements of the system, define the boundaries of version v1 (MVP), and provide a basis for software design documentation (SDD) and testing.

1.2. System Purpose

The system is a web application that allows a registered user to maintain a personal catalog of media content — books, TV series, and films. The application provides tracking of viewed and read content: recording statuses, ratings, notes, tags, repeated viewings (Sessions), organizing items into collections, and viewing personal statistics.

1.3. Document Context and Audience

This document was prepared as part of the “Information Systems Design” course (SDD cycle). Target audience: the course instructor and system developers. A reader of this document should obtain sufficient information to develop the system architecture and testing plan without additional clarification.

1.4. Document Structure

- Section 2 — general system description, MVP boundaries, user roles, operating environment.
- Section 3 — key term definitions.
- Section 5 — complete list of functional requirements grouped by functional blocks.
- Section 6 — non-functional requirements: performance, security, reliability, usability, compatibility, validation.
- Section ?? — acceptance criteria for each requirement.
- Section 8 — BDD scenarios in Gherkin format.
- Section ?? — requirements traceability matrix.
- Appendix A — features outside the MVP scope.

2. General System Description

2.1. Product Overview and Perspective

The system is a standalone web application with no external service integrations within the MVP scope. Each registered user has their own isolated data space — a catalog, collections, and statistics. The system is not part of a larger platform and does not provide a public API in version v1.

2.2. Functional Blocks

The system implements the following functional blocks:

1. Authentication and account management.
2. Personal media content catalog.
3. Content item management (CRUD).
4. Tags and tag-based filtering.
5. Sessions — recording repeated viewings or readings.
6. Collections — custom named groups of items.
7. Automatic completion date fill.
8. Personal statistics.

2.3. User Roles

Version v1 has a single user role. No administrative role exists.

Table 1. System User Roles

Role	Description
User	Registered account owner. Has access exclusively to their own catalog, collections, and statistics. May perform all CRUD operations on their own data.

2.4. Operating Environment

The system is implemented as a web application. A native mobile application is not planned for version v1. The system must function correctly in the latest two versions of Chrome, Firefox, Edge, and Safari browsers, including mobile Safari on iOS. The backend implementation language is not specified by this document and is determined at the architecture design stage.

2.5. Design and Implementation Constraints

- Implementation exclusively as a web application.
- Version v1 supports only one interface language.
- The system does not integrate with external media content databases (TMDB, Google Books, etc.).
- Each user has access exclusively to their own data; cross-user access is not provided.

2.6. MVP Boundaries — What Is Not Included in Version v1

The following features are technically feasible but deliberately excluded from the MVP to constrain the scope of the first release. A detailed list is provided in Appendix A.

3. Glossary

Table 2 lists the definitions of key terms used in this document.

Table 2. System Glossary

Term	Ukrainian equivalent	Definition
Catalog	Каталог	The complete list of all content items added by the user.
Content Item	Елемент контенту	A single entry in the catalog — a book, TV series, or film with its own attributes (type, status, rating, tags, etc.).
Collection	Добірка	A user-created named group of arbitrary catalog items; one item can belong to multiple collections simultaneously.
Session	Session	An individual record of a viewing or reading of an item; contains date, rating, and note; one item can have multiple Sessions.
Tag	Тег	An arbitrary text label that the user attaches to an item for categorization.
Status	Статус	The current state of work on an item: <i>planned / in progress / completed / on hold.</i>

4. Development Methodology

4.1. Specification-Driven Development (SDD)

The system is developed in accordance with the Specification Driven Development (SDD) methodology. The specification is the controlling source of system behavior — no source code may define behavior independently of this document.

Required process for every behavioral change:

1. Update this document before making any code changes.
2. Assign or update a requirement identifier (REQ-F-NNN or REQ-NF-NNN).
3. Define an acceptance criterion (AC-NNN) for each requirement.
4. Write an automated test before implementation (TDD, Section 4).
5. Run the test and confirm it fails for the expected reason (Red).
6. Implement the minimal code that passes the test (Green).
7. Refactor only after tests pass (Refactor).
8. Update log artifacts after completing the iteration.

4.2. Mandatory Test-Driven Development (TDD)

TDD is mandatory for all implementation work. The Red / Green / Refactor cycle applies to every functional change:

- **Red** — write a failing test first.
- **Green** — implement the minimal code that passes the test.

- **Refactor** — improve the code without changing external behavior.

No production code is accepted unless it is covered by at least one automated test. Real external systems (DBMS, third-party APIs) are not used in unit tests — deterministic substitutes (stubs / mocks) are used instead.

4.3. Traceability Rule

Every implemented system behavior must have full traceability in the chain:

Listing 1. Traceability Chain

```
1  REQ -> AC -> Test -> Implementation
```

Functionality without a requirement identifier is not part of the system. A requirement without an acceptance criterion is not ready for implementation.

4.4. Test Log Artifacts

After each test run, the development agent records a log file with the results. The log file must contain:

- total number of tests and the count of passed / failed;
- the status of each individual test (passed / failed / skipped);
- execution time of the test suite.

The log file is a machine-readable verification artifact — based on it, the development agent and human reviewer determine whether the task has been completed according to the specification (the “Is verified?” step in the SDLC process).

5. Functional Requirements

5.1. Authentication and Account Management

Table 3. Authentication and Account Requirements

ID	Requirement
REQ-F-001	The system shall allow user registration with email, password, and nickname (all three fields are mandatory). Re-registration with the same email is not allowed.
REQ-F-002	The system shall allow account login with email and password.
REQ-F-003	The system shall allow account logout. After logout, access to protected pages requires re-authentication.
REQ-F-029	The user may change their nickname in account settings.

5.2. Catalog

Table 4. Catalog Requirements

ID	Requirement
REQ-F-004	The system shall display a personal catalog — a list of all user content items.
REQ-F-005	The system shall provide catalog filtering by item type (book / TV series / film).
REQ-F-006	The system shall provide catalog filtering by item status. Available status values: “planned”, “in progress”, “completed”, “on hold”.
REQ-F-023	The system shall provide catalog item search by title.
REQ-F-026	The system shall provide catalog sorting: by date added (newest / oldest) and by rating (highest to lowest and vice versa).

5.3. Content Item Management

Table 5. Content Item Management Requirements

ID	Requirement
REQ-F-007	The system shall allow adding a new content item with attributes: title (required), type (required), status, rating (0–10), notes, tags. The form displays type-specific optional fields (REQ-F-024) and a cover image field (REQ-F-025).
REQ-F-008	The system shall allow editing any attribute of an existing content item.
REQ-F-009	The system shall allow deleting a content item from the catalog. A deleted item is automatically removed from all collections.
REQ-F-024	The add and edit item form shall display type-specific optional fields depending on the selected type (see Table 6).
REQ-F-025	Each content item shall have a cover image: by default — one of three illustrations corresponding to the type. The user may replace it with their own image (optional).
REQ-F-028	Cover image formats: JPEG, PNG, WebP; maximum file size: 5 MB. The system notifies the user when limits are violated.

5.4. Tags

5.5. Sessions (Repeated Viewings / Readings)

5.6. Collections

5.7. Automatic Completion Date Fill

5.8. Personal Statistics

Table 6. Type-Specific Content Item Fields

Type	Optional Fields
Book	Author, number of pages
TV Series	Number of seasons
Film	Director, duration

Table 7. Tag Requirements

ID	Requirement
REQ-F-010	The system shall allow adding arbitrary tags to a content item. When entering a tag, the system offers autocomplete based on the user's previously entered tags. One item can have multiple tags. A tag can be removed at any time.
REQ-F-011	The system shall provide catalog filtering and search by tag.

6. Non-Functional Requirements

6.1. Performance

REQ-NF-001. System pages shall load in an acceptable time for the user under normal network connection conditions.

6.2. Security

REQ-NF-002. User passwords are stored exclusively in a protected form — as hashes (e.g., bcrypt). Access to protected resources requires an active authenticated session.

REQ-NF-007. The authorization token is stored in an HTTP-only cookie to prevent XSS attacks. All requests that modify system state must be protected against CSRF attacks.

6.3. Reliability

REQ-NF-003. All user data is stored in the database. No user action shall result in the loss of previously entered data without explicit deletion confirmation from the user.

6.4. Usability

REQ-NF-004. The system's interface shall be responsive for desktop and mobile browsers. Separate tablet optimization is not planned for version v1.

REQ-NF-008. When the catalog is empty, a collection is empty, or search / filter results are absent, the system shall display an informative message instead of an empty list.

6.5. Compatibility

REQ-NF-005. The system shall function correctly in the latest two versions of: Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari (including mobile Safari on iOS).

Table 8. Session Requirements

ID	Requirement
REQ-F-012	The system shall allow recording multiple Sessions for a single content item; each Session has its own date, rating, and note.
REQ-F-013	The system shall display the list of all Sessions of a content item in reverse chronological order (newest to oldest), with the total count.
REQ-F-014	The system shall allow editing and deleting an individual Session.

Table 9. Collection Requirements

ID	Requirement
REQ-F-015	The system shall allow adding a single catalog item to multiple collections simultaneously (many-to-many relationship).
REQ-F-016	The system shall allow creating a new named collection.
REQ-F-017	The system shall allow editing a collection name.
REQ-F-018	The system shall allow deleting a collection. Deleting a collection does not delete items from the catalog.
REQ-F-019	The system shall allow removing an item from a collection without deleting it from the catalog or other collections.
REQ-F-020	The system shall display the contents of a collection — the list of items it contains, with the name, type, and status of each.
REQ-F-027	The system shall provide sorting of items within a collection: by date added (newest / oldest) and by rating (highest to lowest and vice versa).

6.6. Input Validation

REQ-NF-006. During new user registration, the system shall validate:

- email — format per RFC 5322;
- password — minimum length of 8 characters;
- nickname — 2 to 50 characters.

If any constraint is violated, the system displays a specific error message indicating the reason for rejection.

6.7. Development Process

REQ-NF-009. After each test run, the development agent shall write a log file to the `logs/` directory. The log file must contain: total number of tests, passed / failed / skipped count, status of each individual test (name + result), and execution time of the test suite.

7. Acceptance Criteria

Table 10. Completion Date Requirement

ID	Requirement
REQ-F-021	When changing an item's status to "completed", the completion date field shall be automatically filled with the current date if no date was entered manually.

Table 11. Personal Statistics Requirements

ID	Requirement
REQ-F-022	The system shall display personal statistics: the number of items with "completed" status broken down by type (books / TV series / films) and by selected time period.
REQ-F-022a	Time period selection for statistics — from predefined options: "this month", "this year", "all time".

Table 12. Requirement Acceptance Criteria

AC ID	Req	Acceptance Criterion
AC-001	REQ-F-001	After filling in a valid email, password, and nickname and confirming registration, the user gains access to an empty personal catalog. Re-registration with the same email is not possible: the system displays a message about an existing account.
AC-002	REQ-F-002	After entering the correct email and password, the user is redirected to their catalog. With incorrect credentials — an error message is displayed and login does not proceed.
AC-003	REQ-F-003	After logout, accessing protected pages redirects to the login page. Re-authentication restores access.
AC-004	REQ-F-004	The catalog displays all added items; for each item, the title, type, status, and rating are visible.
AC-005	REQ-F-005	When selecting type "Books", only items of type "book" are displayed; when the filter is cleared — all items are shown again.
AC-006	REQ-F-006	When selecting status "completed", only items with that status are displayed; when cleared — all items are shown again.
AC-007	REQ-F-007	After filling in required fields (title, type), the item appears in the catalog with its corresponding attributes and cover image.
AC-008	REQ-F-008	After editing and saving, the updated attributes are shown in the catalog and on the item page.
AC-009	REQ-F-009	After confirming deletion, the item disappears from the catalog and from all collections it belonged to.

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AC ID	Req	Acceptance Criterion
AC-010	REQ-F-010	An added tag appears on the item card. When typing in the tag field — autocomplete suggestions from previously used tags are shown. Removing a tag does not affect other attributes.
AC-011	REQ-F-011	When filtering by tag, only items with that tag are displayed; the result updates when the filter is changed or cleared.
AC-012	REQ-F-012	For an item with two or more Sessions — a list of all records with date, rating, and note is shown.
AC-013	REQ-F-013	Sessions are sorted from newest to oldest; the total count is indicated.
AC-014	REQ-F-014	After editing a Session — updated data is shown. After deletion — the session disappears and the count decreases.
AC-015	REQ-F-015	An item is simultaneously displayed in each collection it has been added to. Removal from one collection does not affect the others.
AC-016	REQ-F-016	After entering a name, the new empty collection appears in the collections list.
AC-017	REQ-F-017	After saving, the new name is displayed in the list and in the page heading.
AC-018	REQ-F-018	After confirming deletion, the collection disappears; items remain in the catalog.
AC-019	REQ-F-019	After removing an item from a collection, it remains in the catalog and in other collections.
AC-020	REQ-F-020	The collection page displays all items with title, type, and status.
AC-021	REQ-F-021	If no date is entered when changing status to “completed” — the current date is automatically set. If entered manually — the entered value is saved.
AC-022	REQ-F-022	The statistics page displays the count of completed items by type and by the selected time period.
AC-023	REQ-F-023	When entering part of a title, the catalog displays only matching items; when cleared — all items are shown.
AC-024	REQ-F-024	The form displays type-specific fields only after a type is selected; all fields are optional.
AC-025	REQ-F-025	Without a custom image — the default illustration for the type is shown. After upload — the custom image is shown; it can be removed to restore the default.
AC-026	REQ-F-026	After selecting a sort parameter, the catalog is reordered; changing the parameter reorders it again.
AC-027	REQ-F-027	After selecting a sort parameter within a collection, the list is reordered.

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AC ID	Req	Acceptance Criterion
AC-028	REQ-F-028	An attempt to upload a file of an unsupported format or larger than 5 MB — an error message is shown and the file is not saved.
AC-029	REQ-F-029	After changing the nickname in account settings, the new name is displayed in the interface.
AC-030	REQ-F-022a	The statistics page contains a time period selector: “this month”, “this year”, “all time”.
AC-031	REQ-NF-006	When a password shorter than 8 characters is entered — a specific message is shown. With an invalid email or a nickname outside 2–50 characters — a corresponding message is shown.
AC-032	REQ-NF-008	When the catalog is empty, a collection is empty, or search results are absent, an informative message is displayed (not an empty list).
AC-033	REQ-NF-009	After running tests, a file with the results of the last run exists in the <code>logs/</code> directory. The file contains: total number of tests, count of passed / failed / skipped, status of each test (name + result), and execution time of the suite.

8. BDD Scenarios

Below are BDD scenarios in Gherkin format for key system functions.

Listing 2. New User Registration

```

1 Scenario: New user registration
2   Given the registration page is open
3   And email "test@example.com" is not yet registered in the system
4   When the user enters email "test@example.com", a password, and a
    ↪   nickname
5     and confirms registration
6   Then the user gains access to an empty personal catalog
7   And a repeated registration attempt with the same email shows the
    ↪   message
8     "An account with this email already exists"
```

9. Requirements Traceability Matrix

Listing 3. Account Login

```
1 Scenario: Login with correct credentials
2   Given the user is registered in the system
3   When they enter the correct email and password and click "Sign in"
4   Then they are redirected to their personal catalog
5   And they see their previously added items
```

Listing 4. Repeated Viewing with a Different Rating

```
1 Scenario: Re-watching a film with a different rating
2   Given content item "Film X" already has one Session with rating 9
3   When the user adds a new Session for the same item
4     with rating 5 and note "didn't enjoy it the second time"
5   Then the content item has two Sessions, each with its own date,
6     rating, and note
7   And both ratings are stored separately, neither overwrites the other
```

Listing 5. Item in Multiple Collections

```
1 Scenario: An item belongs to multiple collections simultaneously
2   Given the user has item "Book Y" in the catalog
3   And collections "Favourites" and "2026" exist
4   When the user adds "Book Y" to both collections
5   Then "Book Y" is displayed in the contents of both collections
6   And removing "Book Y" from collection "2026" does not remove it
7     from "Favourites" or from the catalog
```

Listing 6. Automatic Completion Date Fill

```
1 Scenario: Automatic completion date fill
2   Given the user is editing an item with status "in progress"
3   When the user changes the status to "completed"
4     and does not enter a date manually
5   Then the system automatically sets the completion date
6     equal to the current date
```

Listing 7. Filtering by Tag

```
1 Scenario: Filtering the catalog by tag
2   Given the catalog contains three items: two tagged "fantasy" and one
3     ↪ without
4   When the user selects the filter by tag "fantasy"
5   Then only the two items tagged with that tag are displayed
6   And when the filter is cleared, all three items are displayed again
```

Table 13. Traceability Matrix

REQ ID	AC ID	BDD	Note
REQ-F-001	AC-001	Listing 2	Registration
REQ-F-002	AC-002	Listing 3	Login
REQ-F-003	AC-003	—	Logout
REQ-F-004	AC-004	—	Catalog
REQ-F-005	AC-005	—	Filter by type
REQ-F-006	AC-006	—	Filter by status
REQ-F-007	AC-007	—	Add item
REQ-F-008	AC-008	—	Edit item
REQ-F-009	AC-009	—	Delete item
REQ-F-010	AC-010	—	Tags with autocomplete
REQ-F-011	AC-011	Listing 7	Filter by tag
REQ-F-012	AC-012	Listing 4	Sessions
REQ-F-013	AC-013	—	View Sessions
REQ-F-014	AC-014	—	Edit / delete Session
REQ-F-015	AC-015	Listing 5	Many-to-many collections
REQ-F-016	AC-016	—	Create collection
REQ-F-017	AC-017	—	Edit collection
REQ-F-018	AC-018	—	Delete collection
REQ-F-019	AC-019	Listing 5	Remove from collection
REQ-F-020	AC-020	—	View collection
REQ-F-021	AC-021	Listing 6	Auto completion date
REQ-F-022	AC-022	—	Statistics
REQ-F-022a	AC-030	—	Statistics time period
REQ-F-023	AC-023	—	Search by title
REQ-F-024	AC-024	—	Type-specific fields
REQ-F-025	AC-025	—	Cover image
REQ-F-026	AC-026	—	Catalog sorting
REQ-F-027	AC-027	—	Collection sorting
REQ-F-028	AC-028	—	Cover image constraints
REQ-F-029	AC-029	—	Profile editing
REQ-NF-001	—	—	Performance
REQ-NF-002	—	—	Security / hashing
REQ-NF-003	—	—	Reliability
REQ-NF-004	—	—	Responsive interface
REQ-NF-005	—	—	Browser compatibility
REQ-NF-006	AC-031	—	Input validation
REQ-NF-007	—	—	Session security (CSRF, HTTP-only)
REQ-NF-008	AC-032	—	Empty states
REQ-NF-009	AC-033	—	Test log files

Appendix A

Features Outside the MVP Scope

The following features are deliberately excluded from version v1 to constrain the scope of the first release. They may be considered for future versions of the system.

- Offline mode — full operation without an internet connection.
- Push and email notifications.
- Multi-language interface.
- “Friends” feature — mutual subscriptions, shared access to collections.
- Recommendations block or “Popular among users”.
- Integration with external media content databases (TMDB, Google Books, etc.).
- Administrative panel.
- Email confirmation at registration (sending a verification code by email).
- Changing email address and changing password in account settings.
- Account deletion by the user.